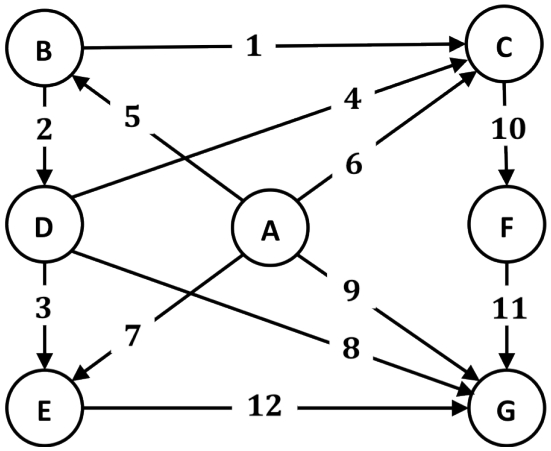
Department of Computer Science
Algorithms Design and Analysis
Spring 2016

Final Exam

Total: _____ **100**

Name:	ID:

1. Tracing Basic Graph Algorithms [/ 25]



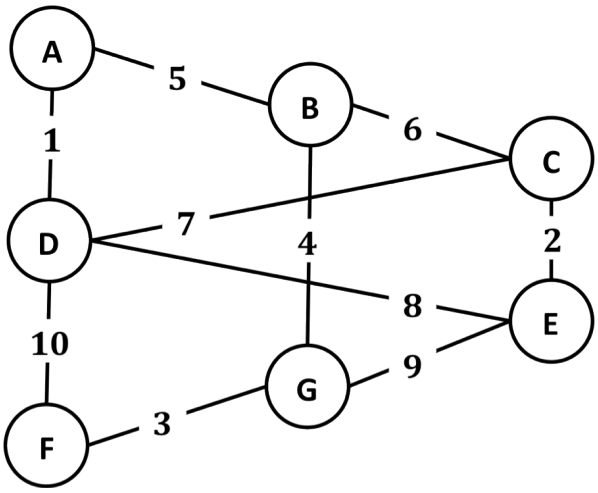
(1) Apply **Depth-First Traversal (DFT)** on the above DAG. Show on the graph the discovery and finish times for each node. Follow the **alphabetical order** if there are ties. [/ 5]

(2) What is the **Topological Order** of the vertices in the above DAG? Assuming that the **DFT**-based algorithm has been used and that **an alphabetical order** has been followed during the DFT traversal. [/5]

1st Vertex	2nd Vertex	3rd Vertex	4th Vertex	5th Vertex	6th Vertex	7th Vertex

(3) Assume that we apply **Dijkstra's algorithm** to find the shortest path from **A** to all the other vertices in the graph. In what order will the vertices be de-queued from the priority queue? [/5]

1st Vertex	2nd Vertex	3rd Vertex	4th Vertex	5th Vertex	6th Vertex	7th Vertex



(4) In what order will the edges be added to the **Minimum Spanning Tree** if **Kruskal's** algorithm is used on the above graph? [/ 5]

(5) In what order will the edges be added to the **Minimum Spanning Tree** if **Prim's** algorithm is applied on the above graph? Assume that **A** is the source vertex. [/ 5]

2. Algorithm Choice [/ 34]

In the Open Day at PSUT, there will be ***N*** games for the students to play. Each game has a start time ***s*** and a finish time ***f***. Each game also has a prize of ***p*** dollars***.*** Ahmad has only ***H hours*** to spend in the open day.

(1) Ahmad can't play two games at the same time. He is willing to play the maximum number of games possible. Which games should he play during the ***H*** hours? [/ 4]

A. What is the closest problem studied in class to this problem?

B. What algorithm design technique did we use to solve it?

C. What is the worst-case asymptotic complexity for finding the solution using the studied algorithm?

(2) Ahmad is sure to win every game he plays. He wants to win the maximum number of dollars. Which games should Ahmad play during the ***H*** hours? Assume that each game has a duration ***d*** (instead of a starting and a finish time). [/ 4]

A. What is the closest problem studied in class to this problem?

B. What algorithm design technique did we use to solve it?

C. What is the worst-case asymptotic complexity for finding the solution using the studied algorithm?

(3) There are ***C*** cameras at PSUT. The dean of student affairs would like to know which games Ahmad has played and in what order. He asked Master Abdullah to write a program that allows any camera to automatically detect which games Ahmad has played and produce an ordered list as follows: G1, G2, G3, etc.

Unfortunately, each camera has produced a different list. The dean wants Master Abdullah to write a new program that finds the two most similar cameras. The similarity between two cameras is measured by the maximum number of games that were recorded by both cameras in the same relative order. [/ 4]

A. What is the closest problem studied in class to the problem of measuring the similarity between two cameras?

B. What algorithm design technique did we use to solve it?

C. What is the worst-case asymptotic complexity for finding the solution using the studied algorithm?

(4) In one of the games, Ahmad is given board of size ***M*** x ***M*** and a set of ***M*** x ***M*** numbers. His task is to determine if it is possible to place the numbers on the board such that the sum of every row and every column is the same, provided that each number appears only once on the board. [/ 4]

A. What algorithm design technique seems useful for solving this problem?

B. What would the worst-case asymptotic complexity be if you use the design technique you have suggested.

(5) For each of the following problems specify the worst-case asymptotic complexity of the best algorithm studied in class. [/ 18]

A. Finding the shortest path between two nodes in a weighted directed acyclic graph.

B. Checking if a directed graph has a negative cycle or not.

C. Finding the Minimum Spanning Tree in a complete weighted undirected graph.

D. Finding the Minimum Spanning Tree in a sparse undirected graph, if we know that all of the edge weights are identical.

E. Finding the list of adjacent vertices of a given vertex in a graph represented as an adjacency matrix.

F. Deleting an edge e(u, v) between two vertices in a directed graph represented as an adjacency list.

3. Miscellaneous [/ 27]

(1) *"To find the shortest path from **s** to **v**, start at **s** and search for **v** using DFS. If there is more than one edge connected to the current node, choose the edge with the minimum weight that has not been chosen yet. Stop when the destination is found."*

Provide a ***counter example*** that shows how this greedy strategy does not always find the optimal solution.

[/ 5]

(2) *"To compute the Minimum Spanning Tree, perform DFS. If there is more than one edge connected to the current node, choose the edge with the minimum weight that does not create a cycle."*

Provide a ***counter example*** that shows how this greedy strategy does not always find the optimal solution.

[/ 5]

(3) The following pseudo-code describes the optimal substructure of finding the shortest path between two vertices in a DAG. Assume that SP(v) is the shortest path from the source to v and that SP(Source) = 0.

```
SP(v)
      Min = INFINITY
      FOR each edge e(u → v)
            D = SP(u)
            IF (D + weight(u → v) < Min)
                  Min = D
```

Show with an example that there are overlapping sub-problems. [/ 5]

(4) Consider a sorting algorithm that was run for a short period of time and then was stopped before it finished. You are currently looking at the array that was modified by this sorting algorithm. Your task is to guess what the sorting algorithm is.

For each of the following sequences, specify if they could be the result of using Merge Sort, Quick Sort, any of them or none of them. ***Justify your answer***.

(Assume that sorting is done in ascending order and that if it is Merge Sort, then it has done at least one merge operation and if it is Quick Sort, then it has done at least one partition operation). [/ 6]

[1 2 3 4 0 2 1 4]

[0 1 2 3 7 6 5 4]

[7 6 5 4 3 2 1 0]

(5) Consider the following algorithm: [/ 6]

```
Foo(N)
      IF (N == 0)
            Return 1

      Sum = 0
      REPEAT 8 Times
            Sum += Foo(N/2)

      RETURN Sum + FACT(N)
```

Where, **FACT**(N) is a function that runs in O(N).

Describe the running time of the above divide and conquer algorithm using a recurrence equation and then solve the recurrence.

4. Algorithm Design [/ 14]

(1) The **Bottleneck Edge** on a path, is the edge with the maximum weight on that path. [/ 10]

Implement (in pseudo-code) function **`FIND_PATH(v, u, T)`**, which finds any path between ***v*** and ***u***, such that the bottleneck edge on the path is at least ***T.*** If no such path exists, report failure.

What is the **worst-case asymptotic complexity** of the algorithm you have implemented?

(2) Using **FIND_PATH**, describe clearly and precisely how the path with the ***maximum Bottleneck Edge*** can be found between two given vertices in $O(E + V)\log E$. [/ 4]